Practical No. 6

Aim : To draw Data Flow Diagram (DFD) for a given case study using StarUML.

Objective

To model the flow of information between external entities, processes, and data stores of the Library Management System using StarUML.

Software Required

StarUML – UML modeling tool.

Case Study

Library Management System

The Library Management System manages books, members, book issuing, book returning, fines and reports.

Theory

A Data Flow Diagram (DFD) represents how data moves through a system.

The main components of a DFD are:

1. External Entity – An entity that interacts with the system.
2. Process – Performs an operation on the data.
3. Data Store – Stores data for future use.
4. Data Flow – Shows movement of data between components.

Level 0 DFD / Context Diagram

In Level 0 DFD, the entire Library Management System is represented as one process.

External Entities

- Librarian
- Member/Student

Data Flows

Librarian → System

- Book details
- Member details
- Issue/return information

System → Librarian

- Book status
- Member information
- Reports

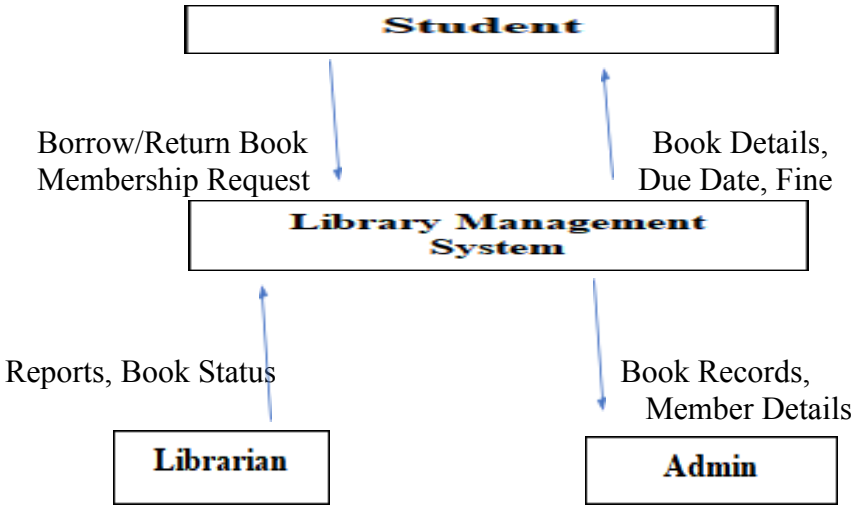
Member → System

- Book search request
- Issue/return request
- Member details

System → Member

- Book availability
- Issue details
- Return details
- Fine information

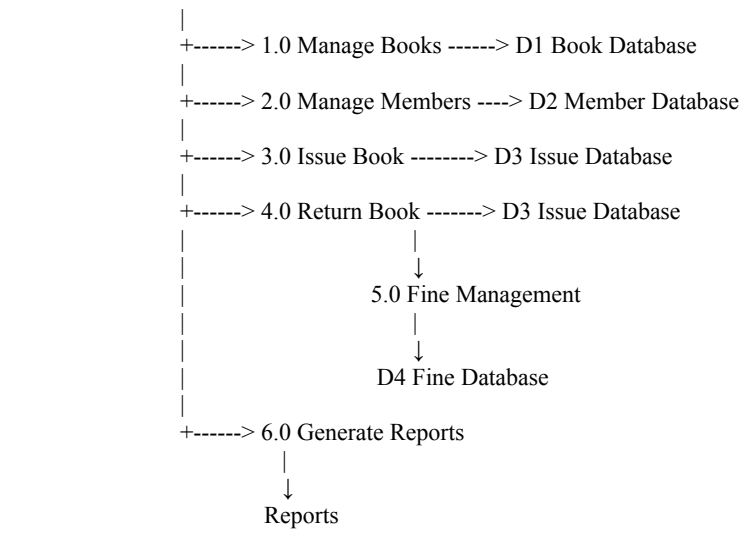
Level 0 DFD



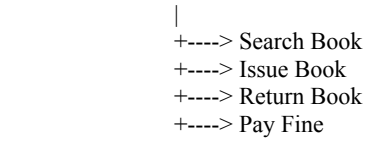
Level 1 DFD
The Library Management System is divided into major processes.
Processes
1.0 Manage Books
2.0 Manage Members
3.0 Issue Book
4.0 Return Book
5.0 Fine Management
6.0 Generate Reports
Data Stores

- D1 – Book Database
- D2 – Member Database
- D3 – Issue/Return Database
- D4 – Fine Database

Level 1 Flow



Member



Procedure to Draw DFD in StarUML
Step 1: Open StarUML

1. Start StarUML.
2. Select File → New.
3. Create a new project.
4. Enter the project name as Library Management System.

Step 2: Create DFD Diagram
StarUML does not provide a dedicated traditional DFD diagram in the same way that specialized DFD tools do. Therefore, you can create the DFD using suitable UML/extension elements or generic diagram elements available in your StarUML installation.

Step 3: Create External Entities
Create entities for:

- Librarian
- Member

Step 4: Create Processes
Add the following processes:

- Manage Books
- Manage Members

- Issue Book
- Return Book
- Fine Management
- Generate Reports

Step 5: Create Data Stores
Represent the following data stores:
- Book Database
- Member Database
- Issue/Return Database
- Fine Database

Step 6: Connect the Components
Use connectors/arrows to show the direction of data flow.
For example:
Librarian → Manage Books → Book Database
Member → Issue Book → Issue Database
Member → Return Book → Issue Database
Return Book → Fine Management → Fine Database

Step 7: Label Data Flows
Label arrows with appropriate data such as:
- Book Details
- Member Details
- Search Request
- Issue Details
- Return Details
- Fine Details
- Reports

Step 8: Save the Diagram
Save the project and export the completed diagram as an image/PDF if required for submission.

Result
The Data Flow Diagram for the Library Management System was successfully designed using StarUML, showing the flow of data between external entities, processes and data stores.

Conclusion
The DFD provides a clear representation of how information enters, is processed, stored and produced by the Library Management System. It helps in understanding the functional flow of the system before implementation.